



Controlled and automatic processes

The way we are able to perform controlled processes (tasks that require our full awareness, such as solving a puzzle) and automatic processes (tasks that require relatively little attention, such as reading a book) is compromised.

Self-control

We may have difficulty controlling our actions and movements, for example walking a straight line while intoxicated. It may also be difficult to restrain emotions, often resulting in outbursts of crying or aggression.

Identifying an altered state

Consciousness is a spectrum from highly alert to total lack of awareness, with a “normal” state somewhere in the middle. Altered states, meanwhile, can be on either side of the scale, with greater or lesser awareness than normal. An altered state can be identified using different criteria.

Level of awareness

In an altered state, our level of awareness of events going on around us—as well as internally—may be increased or decreased compared with normal waking consciousness. More often, our level of awareness is lowered in an altered state.

Emotional awareness

Often in an altered state we will have less emotional awareness (the experience of emotions), as well as finding it difficult to control those emotions. This can make us more or less affectionate, aggressive, or anxious.

Perceptual and cognitive distortions

Perception may be altered. Normal processes for storing and retrieving memories may be more fragmented or less accurate. Thought processes may be disorganized and less logical.

Time orientation

In an altered state, our sense of time (see pp.174–175) can become distorted; time may appear to slow down or speed up. This is because there is less awareness of time passing, just as we are unaware of time while we sleep.



382 DAYS

**THE LONGEST
RECORDED FAST
FROM SOLID FOOD**

Altered states in the brain

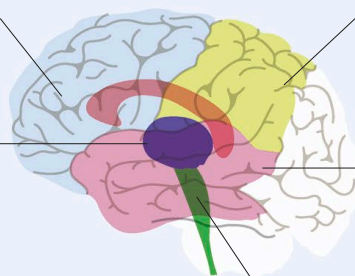
Altered states can lead to a range of experiences, from feelings of bliss to a sense of terror. These experiences are generated by a similarly diverse range of neural activity in various parts of the brain. Alterations to normal brain function can result in our brain distorting incoming information, leading to auditory or visual hallucinations, memory distortion, or delusions.

Decrease in activity in frontal lobe reduces ability to reason and make decisions

Thalamus—which acts as gateway between limbic system and frontal cortex—can be inhibited

Altered activity in parietal lobe distorts spatial judgments and time perception

Changes in temporal lobe function lead to unexplainable experiences such as hallucinations



Locating altered states

In an altered state, activity in different areas of the brain may increase or decrease, distorting how we perceive the world.

Signals from reticular formation, which plays important role in consciousness, can be reduced